# DANNY'S RACE CAR

## ELECTRICAL HARNESS / REV J

L94 / LS3 | Nexus R3 | Delta ABS Gen 2 | IC-7 | 15-button CAN keypad

DESIGN REVIEW DRAFT — OPEN THIS ROOT FILE IN KICAD

02 / BATTERY, ISOLATION AND STARTER

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03 / NEXUS R3 PIN ALLOCATION

File: 03_R3_Pin_Allocation.kicad_sch

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### WHAT IS VERIFIED

R3 and PD16 connector functions; Delta supplied pin allocations; owner equipment choices. Circuit assignments are a draft.

### WHAT REMAINS OPEN

TBD devices, selected ratings, sensor cavity mapping, ABS compatibility, isolation, CAN configuration and physical harness routing.

DRAFT — NOT A BUILD RELEASE

**Danny's Race Car**

Sheet: /

File: Race_Car_RevJ.kicad_sch

***Title: Race car electrical harness — system index***

| Size: A3 | Date: 2026-10-25 | **Rev: J — REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 1/19 |

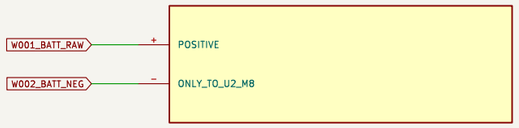
# 02 / BATTERY, ISOLATION AND STARTER

CARTEK GT is the master isolator; K1 removed. F0 feeds TB2 directly. Returns are downstream of isolation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

BT1 | BRAILLE G30H

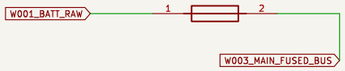
## Battery




PHYSICAL ROUTE: battery-side BATT_RAW block -> F2 -> M1 battery B+ stud -> F3 -> G1 B+.

F0 | MAIN BUS FUSE / SELECT

## MAIN BUS FUSE / SELECT



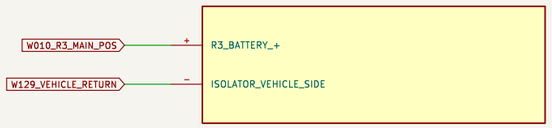

F1 | R3 FEED / SELECT

## Engine controller supply fuse



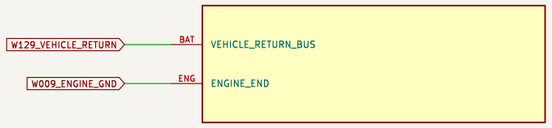

J1 | R3 MAIN SURLOK

## Engine controller power connector




J2 | GROUND STRAP / ENGINE

## Engine ground strap




U2 isolates NEGATIVE. Positive starter/alternator cables remain battery-connected when OFF: insulate, route and protect them.

GT M6_IGN -> S1 -> R3 A13 is the engine-stop control. F0 -> TB2 is permanently battery-positive when connected.

F5 protects the relay-contact/solenoid branch; value TBD. Estimated 30A solenoid pull-in is NOT a measurement. No motor current through U3.

K1 omitted by owner. Verify R3 ignition-OFF stops injection/ignition/fuel and PD16 loads, including CAN-loss/shutdown timing.

No wire except U2 M8 may return to BT1 negative. R3/PD16 and every load use VEHICLE_RETURN; verify with all service cables.

W011: one shared start/charge trunk. F3 branches at M1 BATTERY B+ stud, never SOL or motor-side stud. Size/protection TBD.

## U3 / 24059 BULKHEAD RELAY WIRING

Large PWR <- F5/2; large OUT -> M1 SOL.

Small CMD <- R3 A18/HBO5 (+12V ON).

Small GND -> VEHICLE_RETURN (not case).

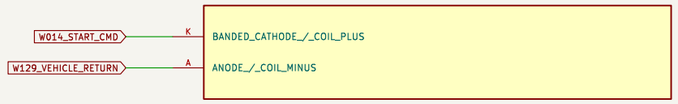
D60 band/K -> CMD; unbanded/A -> GND.

PWR/OUT/CMD/GND are installer tags.

12V coil: 17.5 ohm / ~0.69A calculated.

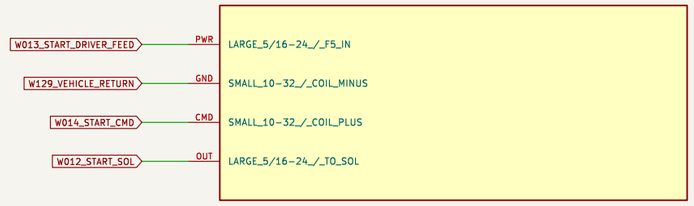
D60 | 1N5408 / BAND TO CMD

## Starter-relay coil suppression diode




U3 | COLE HERSEE 24059 / 12V / 85A

## Cole Hersee 24059 starter relay



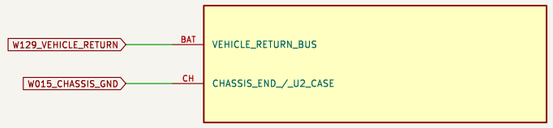

F5 | START DRIVER FUSE / SELECT

## START DRIVER FUSE / SELECT



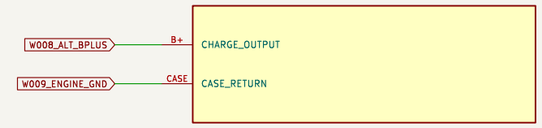

J3 | GROUND STRAP / CHASSIS

## Chassis ground strap




G1 | 79376G / ONE-WIRE

## Alternator



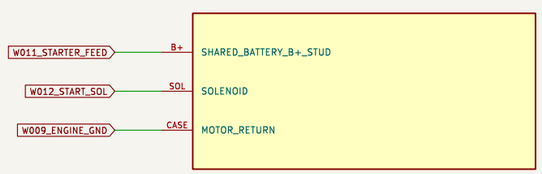

F3 | ALT BRANCH FUSE / AT STARTER

## ALT BRANCH FUSE / AT STARTER




M1 | VW HIGH-TORQUE STARTER

## Starter motor




F2 | SHARED CABLE FUSE / AT BATTERY

## Shared starter / charging cable fuse




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**Danny's Race Car | Harness design**

Sheet: /02 / BATTERY, ISOLATION AND STARTER/

File: 02_Power.kicad_sch

***Title: 02 / BATTERY, ISOLATION AND STARTER***

| Size: A3 | Date: 2026-10-25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 2/19 |

# 03 / NEXUS R3 PIN ALLOCATION

ONE R3 with owner-selected HT-183200 flying-lead harness. Supplied controller leads need routing and free-end termination.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J10 | R3 A / AMP34 KEY1

## Engine controller — connector A

| | | |
|---|---|---|
| W016_INJ_1 | 1 | INJECTOR_1 |
| W017_INJ_2 | 2 | INJECTOR_2 |
| W018_INJ_3 | 3 | INJECTOR_3 |
| W019_INJ_4 | 4 | INJECTOR_4 |
| W020_INJ_5 | 5 | INJECTOR_5 |
| W021_INJ_6 | 6 | INJECTOR_6 |
| W022_INJ_7 | 7 | INJECTOR_7 |
| W023_INJ_8 | 8 | INJECTOR_8 |
| | 9 | DPO1_/_SPARE |
| W025_R3_SHIELD_RETURN | 10 | GROUND_OUT_/_SHIELDS |
| W026_R3_CAN_RETURN | 11 | GROUND_OUT_/_CAN |
| | 12 | DPO2_/_SPARE |
| W028_R3_WAKE | 13 | IGN_SWITCH_INPUT |
| | 14 | DPO3_/_SPARE |
| | 15 | DPO4_/_SPARE |
| | 16 | DPO5_/_SPARE |
| | 17 | DPO6_/_SPARE |
| W014_START_CMD | 18 | HBO5_/_START_CMD |
| W029_DBW_MA | 19 | HBO1_/_DBW_MOTOR_A |
| W030_DBW_MB | 20 | HBO2_/_DBW_MOTOR_B |
| W130_ABS_IGN_SOURCE | 21 | HBO3_/_ABS_IGNITION |
| W032_COCKPIT_12V | 22 | HBO4_/_COCKPIT_12V |
| W033_CAN1_H | 23 | CAN1_HIGH |
| W034_CAN1_L | 24 | CAN1_LOW |
| W131_PD16_WAKE | 25 | HBO6_/_PD16_WAKE |
| | 26 | IGN_SOURCE_/_UNUSED |
| W036_IGN_1 | 27 | IGNITION_1 |
| W037_IGN_2 | 28 | IGNITION_2 |
| W038_IGN_3 | 29 | IGNITION_3 |
| W039_IGN_4 | 30 | IGNITION_4 |
| W040_IGN_5 | 31 | IGNITION_5 |
| W041_IGN_6 | 32 | IGNITION_6 |
| W042_IGN_7 | 33 | IGNITION_7 |
| W043_IGN_8 | 34 | IGNITION_8 |

J11 | R3 C / AMP34 KEY2

## Engine controller — connector C

| | | |
|---|---|---|
| W044_CRANK_SIG | 1 | TRIGGER_+ |
| | 2 | TRIGGER_-_/_HOLD |
| W045_CAM_SIG | 3 | HOME_+ |
| | 4 | HOME_-_/_HOLD |
| | 5 | SPI1_/_SPARE_-_NO_CLUTCH_SWITCH |
| W047_BRAKE_REQUEST | 6 | SPI2_/_BRAKE_REQUEST |
| | 7 | SPI3_/_SPARE |
| | 8 | SPI4_/_SPARE |
| | 9 | SENSOR_8V_/_SPARE |
| W048_APP1 | 10 | AVI1_/_APP1 |
| W049_APP2 | 11 | AVI2_/_APP2 |
| W050_TPS1 | 12 | AVI3_/_TPS1 |
| W051_TPS2 | 13 | AVI4_/_TPS2 |
| W052_OIL_PRESS | 14 | AVI5_/_OIL_PRESS |
| W053_FUEL_PRESS | 15 | AVI6_/_FUEL_PRESS |
| W054_COOLANT_TEMP | 16 | AVI7_/_COOLANT_TEMP |
| W055_AIR_TEMP | 17 | AVI8_/_AIR_TEMP |
| | 18 | AVI9_/_SPARE_-_ONBOARD_MAP |
| | 19 | SPI5_/_SPARE |
| | 20 | SPI6_/_SPARE |
| W057_CAN2_H | 21 | CAN2_HIGH |
| W058_CAN2_L | 22 | CAN2_LOW |
| W059_KNOCK1 | 23 | KNOCK1 |
| W060_KNOCK2 | 24 | KNOCK2 |
| W061_R3_REF5 | 25 | SENSOR_5V_/_200mA |
| W062_R3_SG | 26 | SIGNAL_GROUND |
| W063_GEAR_SIG | 27 | AVI10_/_GEAR |
| W123_SHIFT_FORCE_SIG | 28 | AVI11_/_SHIFT_FORCE |
| W064_WBO_HTR_PLUS | 29 | WBI1_/_HTR_PLUS |
| W065_WBO_INPUT | 30 | WBI1_/_INPUT |
| W066_WBO_PUMP | 31 | WBI1_/_PUMP |
| W067_WBO_NERNST | 32 | WBI1_/_NERNST |
| W068_WBO_HTR_MINUS | 33 | WBI1_/_HTR_MINUS |
| W069_WBO_CAL | 34 | WBI1_/_CAL |

One wideband: onboard WBI1 / C29-C34. No external WB controller. Confirm sensor type/adapter; configure NSP.

A10/A11 are ground OUTPUTS: do not bond them directly to battery, engine or chassis. C26 is sensor return only.

DBW HBO1/2; HBO3 = ABS ignition; HBO4 = cockpit; HBO6 = PD16 wake. Separate protected feeds; verify each load.

HT-183200 supplies labeled flying leads, CAN plugs and wideband plug. Keep shielding; verify revision, reach and sensor adapters.

SPI3-6 remain spare; AVI11/C28 = HGT. A26 is NC: CARTEK U2 M6_IGN is the only source for S1 / R3 A13 (sheet 18).

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**Danny's Race Car | Harness design**

Sheet: /03 / NEXUS R3 PIN ALLOCATION/
File: 03_R3_Pin_Allocation.kicad_sch

***Title: 03 / NEXUS R3 PIN ALLOCATION***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-25 | **Rev: J — REVIEW** |
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 3/19 |

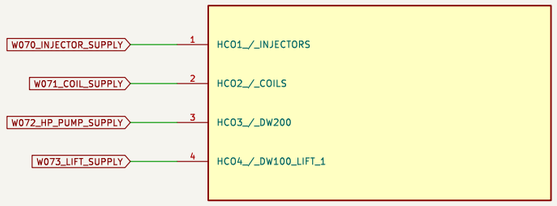
# 04 / ENGINE POWER, INJECTION AND IGNITION

2011 L94 with LS3 cam/intake/throttle body | Donor connectors reused only after condition and cavity checks.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J12 | R3 E / DTP4

## Haltech R3 – integrated PDM outputs




### ELECTRONIC OUTPUT PROTECTION / NO DOWNSTREAM BANK FUSES

HCO1 –> all injector +12V branches; HCO2 –> all coil +12V branches.

Set fuse current / startup time / delay / retries in NSP.

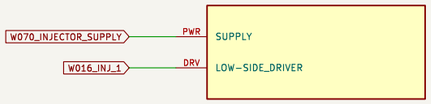
Each channel senses the SUM of its connected loads.

All branch wires and contacts must tolerate the programmed trip curve.

25A is output capacity, NOT a selected protection setting.

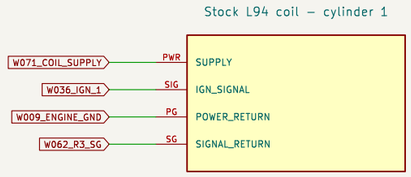
INJ1 | INJECTOR / CYL 1

## LS3 injector – cylinder 1



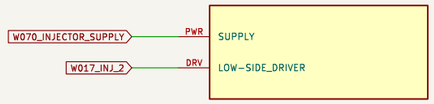

IC1 | COIL / CYL 1

## Stock L94 coil – cylinder 1



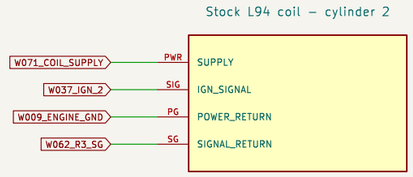

INJ2 | INJECTOR / CYL 2

## LS3 injector – cylinder 2



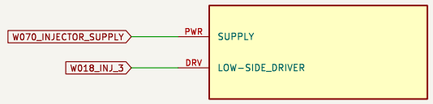

IC2 | COIL / CYL 2

## Stock L94 coil – cylinder 2



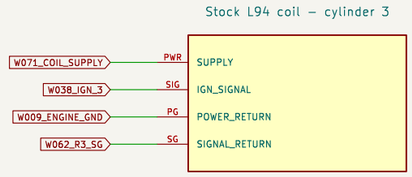

INJ3 | INJECTOR / CYL 3

## LS3 injector – cylinder 3



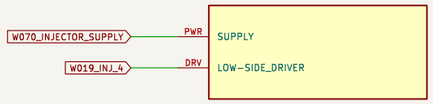

IC3 | COIL / CYL 3

## Stock L94 coil – cylinder 3



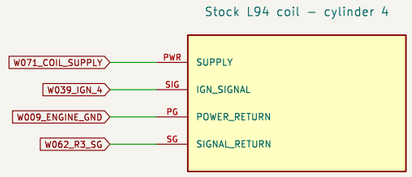

INJ4 | INJECTOR / CYL 4

## LS3 injector – cylinder 4



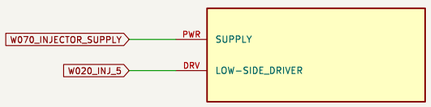

IC4 | COIL / CYL 4

## Stock L94 coil – cylinder 4



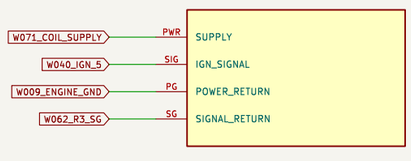

INJ5 | INJECTOR / CYL 5

## LS3 injector – cylinder 5



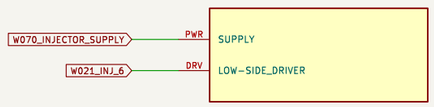

IC5 | COIL / CYL 5

## Stock L94 coil – cylinder 5



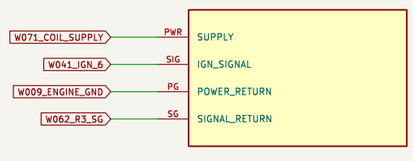

INJ6 | INJECTOR / CYL 6

## LS3 injector – cylinder 6



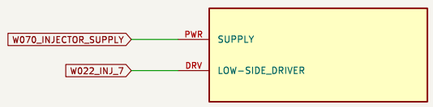

IC6 | COIL / CYL 6

## Stock L94 coil – cylinder 6



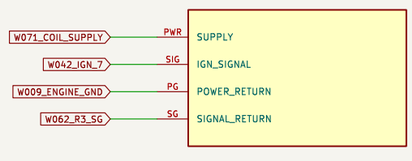

INJ7 | INJECTOR / CYL 7

## LS3 injector – cylinder 7



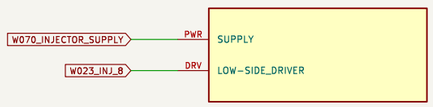

IC7 | COIL / CYL 7

## Stock L94 coil – cylinder 7



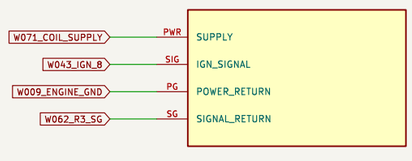

INJ8 | INJECTOR / CYL 8

## LS3 injector – cylinder 8




IC8 | COIL / CYL 8

## Stock L94 coil – cylinder 8




HOLD P02: HCO1/2 must protect retained OEM wire/contact branches. Stock L94 coil and injector wiring is retained; see sheet 19.

PWR/DRV/SIG/PG/SG remain functional device IDs. Stock bank connector reference cavities are on sheet 19; inspect donor before wiring.

IGN and INJ channel number = cylinder number; firing order is an ECU configuration item, not a reason to reorder wires.

Do not merge HCO1/HCO2 through donor shared feeds. Coil low-reference grounding remains HOLD pending exact coil/R3 evidence.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /04 / ENGINE POWER, INJECTION AND IGNITION/

File: 04_Engine_Power.kicad_sch

***Title: 04 / ENGINE POWER, INJECTION AND IGNITION***

| Size: A3 | Date: 2026-10-25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 4/19 |

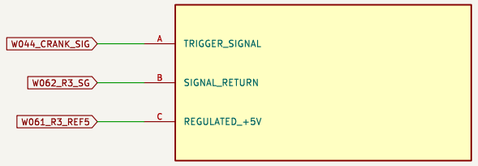
# 05 / ENGINE SENSORS AND WIDEBAND

R3 analog allocation is on sheet 03. Sensor model, reference voltage and calibration must match before connection.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

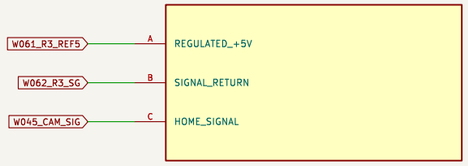
B1 | CRANK / GM 12703627

## Crankshaft position sensor




B2 | CAM / GM 12591720

## Camshaft position sensor




MAP SENSOR IS INSIDE NEXUS R3

Onboard 4-bar absolute MAP selected in NSP.

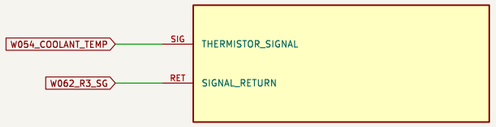
Intake manifold -> pressure hose -> R3 MAP barb.

Pneumatic connection only; no external sensor wires.

AVI9 / C18 spare (onboard MAP). One wideband only.

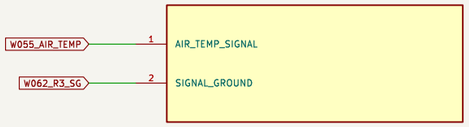
B4 | COOLANT TEMP

## Coolant temperature sensor



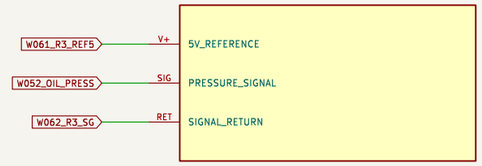

B5 | HT-010206 / INTAKE AIR

## Haltech HT-010206 intake air temperature



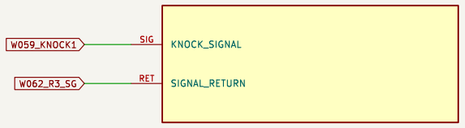

B6 | OIL PRESSURE / TBD

## Oil pressure sensor



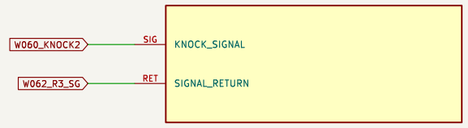

B7 | KNOCK / BANK 1

## Knock sensor - bank 1



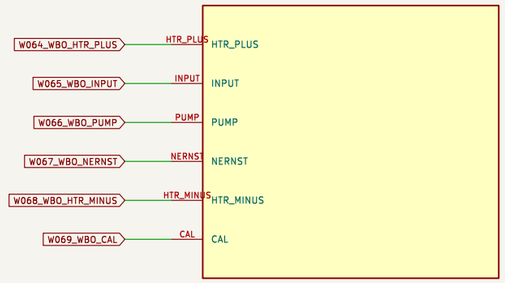

B8 | KNOCK / BANK 2

## Knock sensor - bank 2




B9 | ONE WIDEBAND / R3 WBI1

## Single wideband O2 / R3 onboard




Crank A -> C1; B -> C26; C -> C25. Cam A -> C25; B -> C26; C -> C3. W078/W079 retired; never reuse.

Shield crank/cam/knock cables; shield/drain to R3 A10 only at ECU end. Sensor return remains C26.

Owner confirms crank GM12703627 and cam GM12591720. Verify molded cavities; C25 200mA includes BOTH sensors.

WBI1 C29-C34: HTR+ / INPUT / PUMP / NERNST / HTR- / CAL. Retain matching sensor/adapter; no separate power/ground.

HT-010206: C17, 1k pull-up, GM Air calibration. O2: select LSU4.9 or NTK in NSP; exact pack/cavities and exhaust location TBD.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /05 / ENGINE SENSORS AND WIDEBAND/

File: 05_Engine_Sensors.kicad_sch

***Title: 05 / ENGINE SENSORS AND WIDEBAND***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 5/19 |

# 06 / DRIVE-BY-WIRE

Pedal GM 19417903 | Stock LS3 throttle body | Function labels below are NOT physical cavity numbers.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

B10 | C6 PEDAL / GM 19417903

## Accelerator pedal

| | | |
|---|---|---|
| W061_R3_REF5 | REF1 | APP1_REFERENCE |
| W048_APP1 | SIG1 | PEDAL_TRACK_1 |
| W062_R3_SG | RET1 | APP1_RETURN |
| W061_R3_REF5 | REF2 | APP2_REFERENCE |
| W049_APP2 | SIG2 | PEDAL_TRACK_2 |
| W062_R3_SG | RET2 | APP2_RETURN |

M2 | LS3 THROTTLE BODY

## Electronic throttle body

| | | |
|---|---|---|
| W029_DBW_MA | MA | MOTOR_A |
| W030_DBW_MB | MB | MOTOR_B |
| W061_R3_REF5 | REF | TPS_REFERENCE |
| W050_TPS1 | TPS1 | POSITION_TRACK_1 |
| W051_TPS2 | TPS2 | POSITION_TRACK_2 |
| W062_R3_SG | RET | TPS_RETURN |

S2 REMOVED / NO CLUTCH SWITCH

R3 C5 / SPI1 is spare and unconnected; W046 retired.

Fresh start request, engine-not-running and bounded crank remain required.

No clutch/neutral switch input; physical master stop remains independent.

HOLD D01: verify all six pedal cavities and all throttle-body cavities, including motor polarity, before terminating.

Pedal pigtail missing: donor connector may be reusable; CPP-LSAPP is a listed candidate. Housing fit does not prove wiring.

APP1/APP2 -> C10/C11; TPS1/TPS2 -> C12/C13. No pull-ups assumed for powered position signals.

Complete Haltech DBW calibration, track correlation, brake override and fault checks before any powered vehicle test.

Owner removes S2 and its two wires. No clutch/neutral start interlock is drawn; SPI1 must not be jumpered.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /06 / DRIVE-BY-WIRE/

File: 06_DBW.kicad_sch

***Title: 06 / DRIVE-BY-WIRE***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 6/19 |

# 07 / FUEL PUMPS, SENDERS AND COOLING

Two DW100 lift pumps -> surge tank -> DW200 feed pump | PROFORM 67049 kit: two 67048 brushless puller fans; 10 A each; independent on/off feeds.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

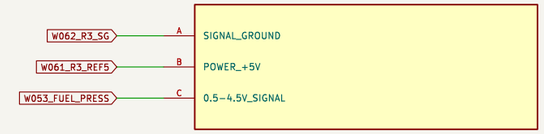
LIFT PUMPS: SEPARATE PROTECTED CHANNELS

Lift 1 = R3 E4; lift 2 = PD16 B4. No F31/F32 inline fuses.

Set each trip curve for its own leads; verify cold/hot startup.

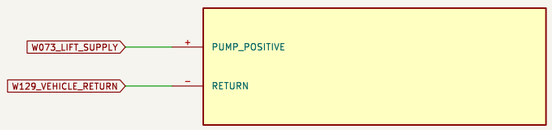
B11 | HT-010910 / 150 PSI FUEL

Haltech HT-010910 fuel pressure



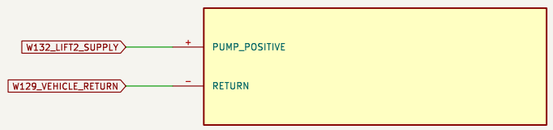

M3 | DW100 / LIFT 1

Fuel lift pump 1



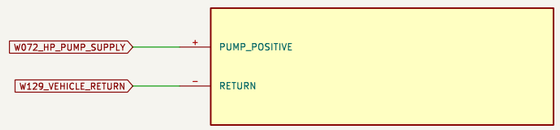

M4 | DW100 / LIFT 2

Fuel lift pump 2



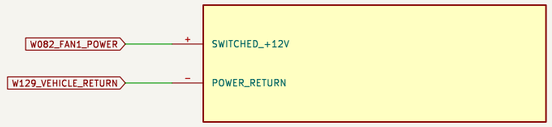

M5 | DW200 / HIGH PRESSURE

High-pressure fuel pump



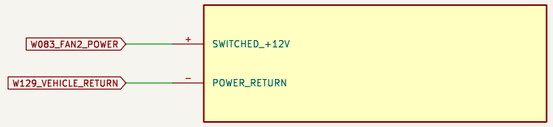

M6 | PROFORM 67048 / 10 A

Radiator fan 1



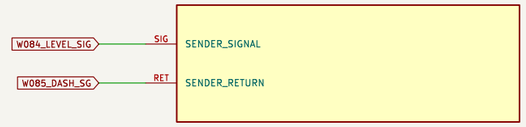

M7 | PROFORM 67048 / 10 A

Radiator fan 2




B12 | FUEL LEVEL / SN38T10

Fuel level sender




HT-010910: A ground / B +5V / C signal -> R3 AVI6 C15. Pull-up OFF; 0.5V = 0 PSI / 4.5V = 150 PSI.

Remote-mount HT-010910 with flexible fuel-rated hose; do not mount directly on rail/block. PROFORM pair: 20 A listed.

Lift pumps now have independent 25A-capable outputs. Select lower trip thresholds from measured load and lead/contact ratings.

Fuel permission requires engine run or timed prime AND valid PD16 communication/output status. Trip/CAN loss -> fuel/engine stop.

Fuel pumps: prime/run logic with engine-stall shutdown; service prime must be momentary/time-limited and master-stop subordinate.

Fans: PD16 B1/B2 supply independent switched +12V (100% when on). No motor-power PWM or external speed-control signal.

Fans: 11.43 x 10.75 x 3.25 in each; pair 22.86 x 10.75 in on 24 x 12 in core. Rear clearance confirmed; verify brackets/shroud.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /07 / FUEL PUMPS, SENDERS AND COOLING/

File: 07_Fuel_Cooling.kicad_sch

***Title: 07 / FUEL PUMPS, SENDERS AND COOLING***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 7/19 |

# 08 / DELTA ABS GEN 2 MAIN CONNECTOR

Bosch 1928405167 / 38 cavities | Source: owner-supplied SCS Delta Gen 2 Rev.3 allocation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J20 | DELTA HCU / 38 PIN

Anti-lock brake unit — main connector

| | | |
|---|---|---|
| W086_ABS_PUMP_POS | 1 | PUMP_SUPPLY_/_40A_NOTE |
| | 2 | FR_SPEED_OUT_/_UNUSED |
| | 3 | ABS_ACTIVE_/_UNUSED |
| W087_ABS_FR_RET | 4 | FR_SENSOR_RETURN |
| W088_ABS_SLIP_5V | 5 | SLIP_5V_REF |
| W089_ABS_PRESS_5V | 6 | PRESSURE_5V_REF |
| W090_ABS_SLIP_RET | 7 | SLIP_RETURN |
| W091_ABS_FL_RET | 8 | FL_SENSOR_RETURN |
| W092_ABS_SG9 | 9 | SENSOR_RETURN |
| W093_ABS_WARN_SINK | 10 | WARNING_LAMP_SINK |
| | 11 | SPEEDOMETER_/_UNUSED |
| | 12 | RR_SPEED_OUT_/_UNUSED |
| W094_ABS_MAIN_POS | 13 | SUPPLY_/_25A_NOTE |
| W095_ABS_PORT2_L_TBD | 14 | CAN2_LOW_/_HOLD |
| W047_BRAKE_REQUEST | 15 | BRAKE_LIGHT_SINK |
| | 16 | RL_SPEED_OUT_/_UNUSED |
| | 17 | FL_SPEED_OUT_/_UNUSED |
| W096_ABS_RL_RET | 18 | RL_SENSOR_RETURN |
| W097_ABS_FL_SIG | 19 | FL_SENSOR_SIGNAL |
| W098_ABS_PORT1_L_TBD | 20 | CAN1_LOW_/_HOLD |
| | 21 | ANALOG1_/_SPARE |
| W099_ABS_SLIP_SIG | 22 | SLIP_ANALOG_SIGNAL |
| W100_ABS_SG23 | 23 | ANALOG_RETURN |
| | 24 | ANALOG2_/_SPARE |
| W101_ABS_POWER_RETURN | 25 | PUMP_GROUND |
| W102_ABS_PORT2_H_TBD | 26 | CAN2_HIGH_/_HOLD |
| | 27 | ANALOG3_/_SPARE |
| W103_ABS_WAKE | 28 | IGNITION_WAKE |
| W104_ABS_RR_RET | 29 | RR_SENSOR_RETURN |
| | 30 | ANALOG4_/_SPARE |
| W105_ABS_RL_SIG | 31 | RL_SENSOR_SIGNAL |
| W106_ABS_RR_SIG | 32 | RR_SENSOR_SIGNAL |
| W107_ABS_PORT1_H_TBD | 33 | CAN1_HIGH_/_HOLD |
| W108_ABS_FR_SIG | 34 | FR_SENSOR_SIGNAL |
| W109_ABS_ENABLE | 35 | ENABLE_/_GROUND_ON |
| W110_ABS_PRESS_FRONT | 36 | FRONT_PRESSURE |
| W111_ABS_PRESS_REAR | 37 | REAR_PRESSURE |
| W101_ABS_POWER_RETURN | 38 | POWER_GROUND |

F41 | ABS PUMP / VALUE TBD

Anti-lock brake pump fuse

W003_MAIN_FUSED_BUS — 1 — 2 — W086_ABS_PUMP_POS

F42 | ABS MAIN / VALUE TBD

Anti-lock brake control fuse

W003_MAIN_FUSED_BUS — 1 — 2 — W094_ABS_MAIN_POS

F43 | ABS WAKE / VALUE TBD

Anti-lock brake wake supply fuse

W130_ABS_IGN_SOURCE — 1 — 2 — W103_ABS_WAKE

J21 | ABS GROUND / HOLD

Anti-lock brake ground connection

| | | |
|---|---|---|
| W101_ABS_POWER_RETURN | ABS | HARNESS_RING |
| W129_VEHICLE_RETURN | BAT | PROPOSED_BATTERY_END |

Owner retains complete Delta loom and its fuses. Guide: pump40A/solenoid20A/ignition10A; older25A label requires revision check.

J21 records the proposed battery return; HOLD A01: reconcile manual chassis-ground wording with pinout battery-negative wording.

HOLD A02: identify which HCU CAN pair serves public CAN and which serves the supplied IMU/diagnostic harness.

ABS sensor grounds remain within ABS. Main connector unused output pins have NC marks; public-interface detail follows.

Finished Delta branches and IMU/private CAN are supplied, not wires to fabricate. Vehicle power/public-CAN interface only.

DRAFT — NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /08 / DELTA ABS GEN 2 MAIN CONNECTOR/

File: 08_ABS_Main.kicad_sch

***Title: 08 / DELTA ABS GEN 2 MAIN CONNECTOR***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-25 | **Rev: J — REVIEW** |
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 8/19 |

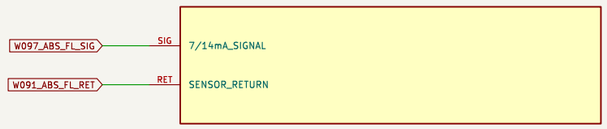
# 09 / ABS SENSORS, CONTROLS AND PUBLIC CAN

Wheel sensors -> Delta ABS -> public CAN -> R3 traction control. Selected path; Gen 2 protocol check remains open.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

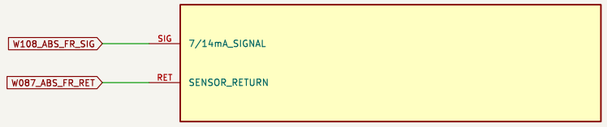
B20 | FRONT LEFT / GM 22742183

Front left wheel-speed sensor




B21 | FRONT RIGHT / GM 22742183

Front right wheel-speed sensor



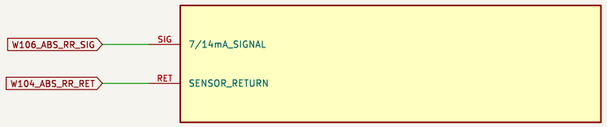

B22 | REAR LEFT / DELTA SENSOR

Rear left wheel-speed sensor



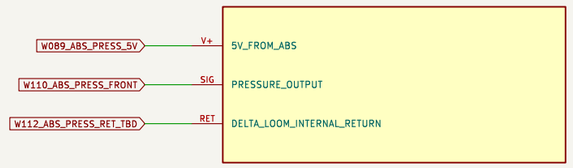

B23 | REAR RIGHT / DELTA SENSOR

Rear right wheel-speed sensor



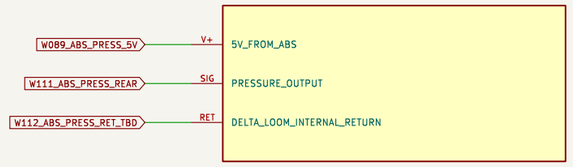

B24 | FRONT BRAKE PRESSURE

Front brake pressure sensor



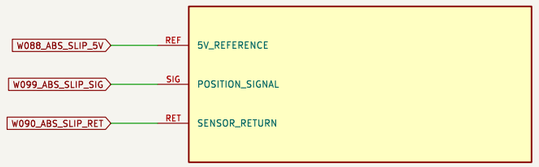

B25 | REAR BRAKE PRESSURE

Rear brake pressure sensor



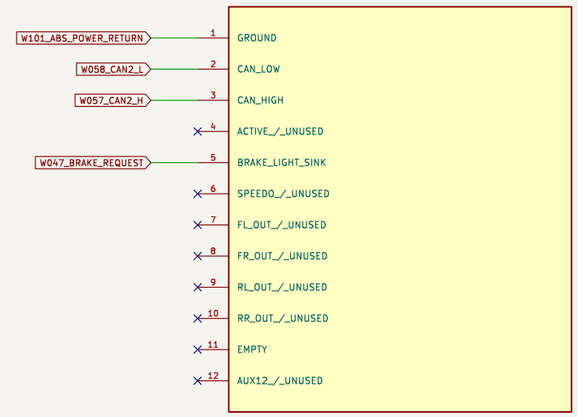

S20 | SLIP SELECTOR / PHYSICAL

Anti-lock brake slip selector



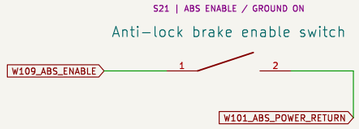

J22 | PUBLIC DTM06-12S

Anti-lock brake vehicle interface



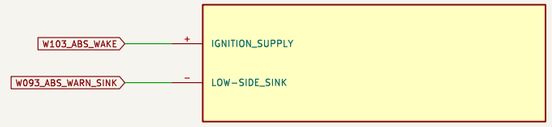

S21 | ABS ENABLE / GROUND ON

Anti-lock brake enable switch




H20 | ABS WARNING LAMP

Anti-lock brake warning lamp




HOLD A03: GM 22742183 signal compatibility/polarity unverified. Check both corners; GM lists it as front-left assembly.

Both pressure sensors use labelled 3-pin Delta loom plugs. Return W112 is supplier-internal; no added ground wire or R3 tap.

IMU/yaw sensor and its private CAN remain in supplied harness: map with verified harness data before build release.

S21/C158 HOLD: current guide describes hardware solenoid-power OFF; old pin35 ground-enable map is not a fabrication instruction.

CAN wheel speeds selected per Haltech support. J22:7-10 unused; verify Gen 2 profile/bitrate (sheet 15). SPI2 = brake request.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /09 / ABS SENSORS, CONTROLS AND PUBLIC CAN/

File: 09_ABS_Sensors.kicad_sch

***Title: 09 / ABS SENSORS, CONTROLS AND PUBLIC CAN***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 9/19 |

# 10 / CAN, DASH, KEYPAD AND WAKE

CAN1: R3 / PD16 / iC-7 / keypad / TMS-4 (1 Mbit/s Haltech). CAN2: Delta ABS public bus, separately verified.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

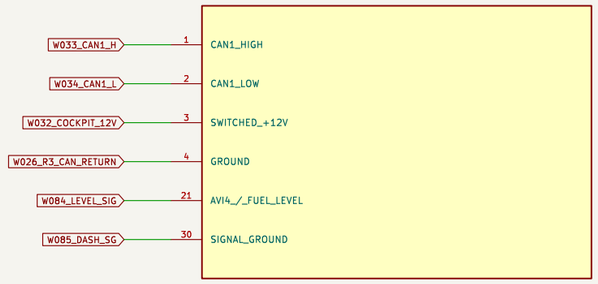
S1 | NORMAL R3 WAKE / U2 PERMIT

Engine controller power switch



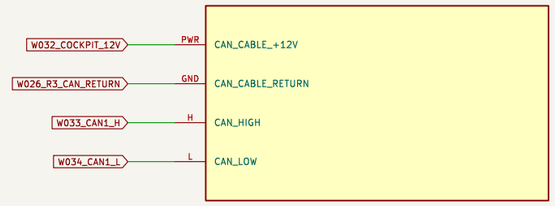

U10 | HALTECH IC-7

Haltech iC-7 digital dash



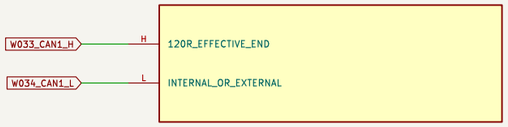

U11 | HT-011502 / 15 BUTTON

Driver control keypad



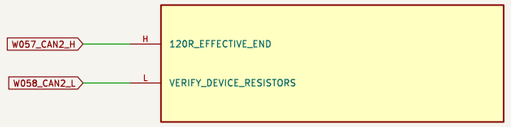

J30 | PD16 INTERNAL / CAN1 END

Dashboard data-bus termination



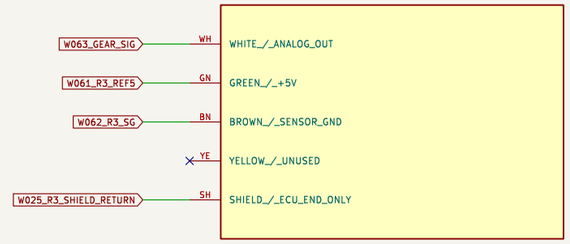

J31 | CAN2 FAR-END TERM

Vehicle data-bus termination




B30 | NOVOTECHNIK RSC-2841-634-211-202

Transaxle gear-position sensor




## KEYPAD / CAMERA CONFIRMED; POSITION 10 PROPOSED

1 Run/Stop  2 Start (momentary)  3 Headlights  4 Tail lights  5 Hazards

6 Fan override  7 Timed fuel prime  8 Dash next  9 Alarm acknowledge  10 Camera ON/OFF

11 Spare  12 Spare  13 Spare  14 Spare  15 Spare

## GEAR DISPLAY – R3 TO IC-7 OVER CAN1

NSP: Gear Position Sensor; AVI10 / C27; pull-up OFF.

Calibrate every forward gear, neutral and reverse in place.

Set non-overlapping voltage windows; check repeatability.

IC-7: display ECU Gear channel. No separate display needed.

S1 is normal wake, supplied only by CARTEK U2 M6_IGN. Emergency stop uses U2/S40 and R3 A13; verify engine/output shutdown and no backfeed.

CAN termination is a topology requirement, not two extra resistors to fit blindly. R3 has selectable internal termination per bus.

CAN1: R3 termination at one end; choose PD16 at the other and enable its termination. J30 represents that same resistor.

Twist each H/L pair; minimize stubs. Do not tie CAN1 to CAN2. R3 A11 total device return current must stay within its rating.

iC-7 fuel level: AVI4 pin 21 / return 30; 240R pull-up for resistive sender. Confirm dash hardware and NSP firmware.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /10 / CAN, DASH, KEYPAD AND WAKE/

File: 10_CAN_Cockpit.kicad_sch

***Title: 10 / CAN, DASH, KEYPAD AND WAKE***

| Size: A3 | Date: 2026-10-25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 10/19 |

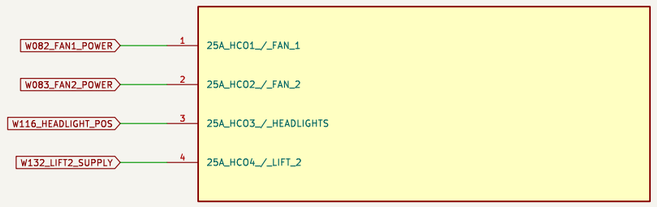
# 11 / AUXILIARY DISTRIBUTION AND LIGHTING

One Haltech PD16 / HT-198100 kit. U20, J40 (sheet 13) and J41 are interfaces on the SAME module.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J41 | PD16 B / DTP4

Haltech PD16 – 25 A power outputs




U20 | PD16 / HT-198100

Haltech PD16 – main power




Control, CAN and all four ground wires: J40 on sheet 13.

Main +12V: 4 AWG manufacturer baseline; F50 rating TBD.

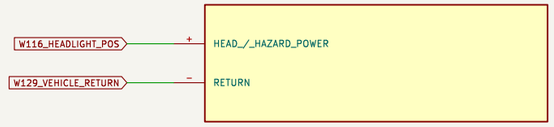
F50 | AUX MAIN / TBD

PD16 main supply fuse



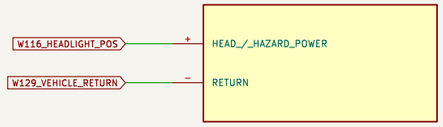

H30 | LEFT HEADLIGHT / TBD

Left LED headlight



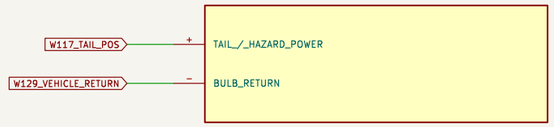

H37 | RIGHT HEADLIGHT / TBD

Right LED headlight



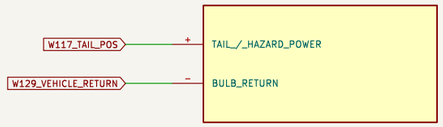

H31 | LEFT TAIL LED / TBD

Left tail LED bulb



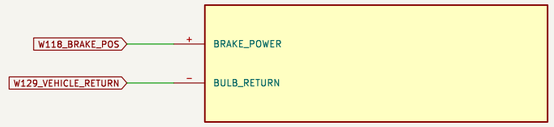

H35 | RIGHT TAIL LED / TBD

Right tail LED bulb



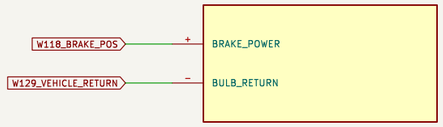

H34 | LEFT BRAKE LED / TBD

Left brake LED bulb




H36 | RIGHT BRAKE LED / TBD

Right brake LED bulb



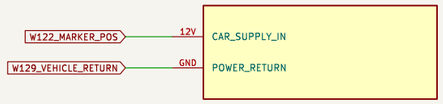

WOLFBOX CAMERA SYSTEM – SHEET 17

A7 / HCO5 -> kit -> USB-C mirror power.

Keypad camera toggle; proposed position 10.

U22 | BRAKE MARKER / CONVERTER

Brake-marker electronics




HEADLIGHT_POS = both headlights / PD16 B3. TAIL_POS = both tail bulbs / A3. BRAKE_POS = both brake bulbs / A4.

Keypad hazards: flash headlights AND tail bulbs together. Hazard logic overrides steady ON so the lamps can turn OFF each cycle.

Brake bulbs stay steady on brake request; no hazard flashing of brake bulbs. A5/HCO3 and A6/HCO4 are now SPARE.

Hazards OFF: restore normal head/tail requests. Implement shared flash phase and verify LED-driver cycling; no lamp models selected.

HOLD L01: configure/test flash logic, CAN-loss behavior, bulb currents, conductor/contact protection and repeated-start allowance.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /11 / AUXILIARY DISTRIBUTION AND LIGHTING/

File: 11_Auxiliary_Power.kicad_sch

***Title: 11 / AUXILIARY DISTRIBUTION AND LIGHTING***

| Size: A3 | Date: 2026-10-25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 11/19 |

# 12 / HARNESS CONVENTIONS AND OPEN ITEMS

This is a connected functional draft for review. It is not a fabrication release or an ECU configuration file.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

## HOW TO READ THIS SET

Identical Wxxx_NAME global labels are electrically connected across sheets.

Wxxx identifies an electrical net; branched nets require individual harness conductor IDs later.

Numeric R3/PD16/Delta/IC-7 pins are documented. Alphabetic functional IDs on TBD parts are not cavity numbers.

NC marks identify deliberately unused pins. A net ending TBD may have only one endpoint.

Reference J10/J11/J12 identifies R3 connectors A/C/E; J1 identifies its main power interface.

## GROUNDING AND CONSTRUCTION

Battery/engine/chassis returns, R3 sensor return, R3 ground outputs and ABS sensor returns are distinct nets.

Do not bond R3 ground-output pins A10/A11 to chassis. Use A10 for shield drains at the ECU end only.

Reuse L94 donor plugs/pigtails after inspection; verify cavity identity and record color changes.

Wire sizes, insulation class, connector contacts, splice sizes and route lengths remain build-stage selections.

Electronic output protection must protect EVERY branch wire/contact; verify trip curves, cranking drop and startup.

## RELEASE HOLDS

P01 CARTEK GT alone selected; verify engine-stop input, return isolation, cutoff timing, charging and starter driver.

E01 Crank/cam PNs identified; verify actual cavities, coil/injector/DBW hardware, 200mA reference and 500mA return budgets.

P02 Electronic trip curves vs weakest injector/coil/pump branches; F01 load/startup, fan settings and hot-idle test.

A01 ABS ground wording; A02 private/public CAN mapping; A03 C7 sensor signal/polarity compatibility.

N01 Gen 2 CAN profile/bitrate, TC loss behavior (sheet 15), PD16 setup; gear/HGT calibration and fuel-level input details.

## CONFIGURATION AND COMMISSIONING

Load/program NSP separately: I/O mapping, output protection, sensor calibration, DBW faults and start permissives.

Check continuity/cavity orientation with electronics unplugged; then check supplies before connecting sensors.

Verify master cutoff with engine running, fuel stall shutdown, fan control and brake-light response.

Review source revisions and actual hardware before termination. No vehicle-powered test was performed.

NASA CCR is the requested safety reference; this drawing does not certify compliance.

Selected CAN traction-control path and spare I/O: sheet 15. Sources and pin/net schedule: README.md / Connections.tsv.

Verified manufacturer electrical limits are ceilings, not chosen fuse-trip settings. No fuse or wire purchase schedule is released.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /12 / HARNESS CONVENTIONS AND OPEN ITEMS/
File: 12_Design_Notes.kicad_sch

***Title: 12 / HARNESS CONVENTIONS AND OPEN ITEMS***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 12/19 |

# 13 / HALTECH PD16 CONNECTIONS AND CONTROL

Owner uses HT-188000 flying-lead harness | PD16 diagram rev 1.5, October 2024 | Peripheral Power Mode / Voltage Sense

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J40 | PD16 A / AMP34 KEY1

## Haltech PD16 – connector A

| | | |
|---|---|---|
| | 1 | HBO1_/_SPARE |
| | 2 | HBO2_/_SPARE |
| W117_TAIL_POS | 3 | 8A_HCO1_/_TAIL_LIGHTS |
| W118_BRAKE_POS | 4 | 8A_HCO2_/_BRAKE_LIGHTS |
| | 5 | 8A_HCO3_/_SPARE |
| | 6 | 8A_HCO4_/_SPARE |
| W121_CAMERA_POS | 7 | 8A_HCO5_/_CAMERA |
| W122_MARKER_POS | 8 | 8A_HCO6_/_BRAKE_MARKER |
| | 9 | 8A_HCO7_/_SPARE |
| | 10 | 8A_HCO8_/_SPARE |
| W124_PD16_SG | 11 | SENSOR_RETURN_/_OIL_+_TRANS |
| | 12 | 5V_REFERENCE_/_UNUSED |
| | 13 | IGN_INPUT_/_UNUSED |
| | 14 | SPI1_/_SPARE |
| | 15 | SPI2_/_SPARE |
| | 16 | SPI3_/_SPARE |
| | 17 | 8A_HCO9_/_SPARE |
| | 18 | 8A_HCO10_/_SPARE |
| | 19 | SPI4_/_SPARE |
| W125_OIL_TEMP_SIG | 20 | AVI1_/_OIL_TEMP |
| W135_TRANS_TEMP_SIG | 21 | AVI2_/_TRANS_TEMP |
| W026_R3_CAN_RETURN | 22 | HALTECH_CAN_GROUND |
| W033_CAN1_H | 23 | HALTECH_CAN_HIGH |
| W034_CAN1_L | 24 | HALTECH_CAN_LOW |
| W131_PD16_WAKE | 25 | CAN_+12V_WAKE_INPUT |
| | 26 | IGN_SOURCE_/_UNUSED |
| | 27 | AVI3_/_SPARE |
| | 28 | AVI4_/_SPARE |
| | 29 | NO_CONNECTION |
| | 30 | NO_CONNECTION |
| W129_VEHICLE_RETURN | 31 | BATTERY_GROUND_1 |
| W129_VEHICLE_RETURN | 32 | BATTERY_GROUND_2 |
| W129_VEHICLE_RETURN | 33 | BATTERY_GROUND_3 |
| W129_VEHICLE_RETURN | 34 | BATTERY_GROUND_4 |

### ONE MODULE – THREE DRAWING REFERENCES

U20 = main positive; J41 = connector B; J40 = connector A.

Peripheral mode: dedicated R3 HBO6/A25 -> PD16 A25.

A13 and A26 are unused. Do not backfeed A25 as an output.

Ground A31–A34 to VEHICLE_RETURN, not BT1 negative.

Use four 18 AWG ground wires per Haltech guide.

A22 is CAN cable ground; not a substitute for A31–A34.

### FAN CONFIGURATION – NO EXTERNAL PWM

Assign PD16 25A HCO1 / HCO2 to separate thermofans.

Use coolant-temperature on/off hysteresis and keypad override.

Temperature thresholds must match the thermostat/cooling test.

Verify power-up, CAN loss, sensor-fault and overcurrent behavior.

Do not make fan-off on track unconditional; retain overheat demand.

Measure hot-idle charging with both fans on; listed pair load 20 A.

CAN hub branch: H/L/return only. Keep PD16 A25 power separate from cockpit +12V; no four-wire power bridge.

Use the manufacturer connector-face view for cavity orientation. A/B prefixes identify connectors, not additional modules.

A3–A10/A17–A18 outputs are limited to 8 A each; B1–B4 to 25 A each. Main connector rating is not an aggregate thermal guarantee.

Configure Peripheral Power Mode, Haltech CAN 1 Mbit/s and PD16 end termination in NSP. Do not add another resistor at J30.

PD16 manufacturer baseline: main 4 AWG, B outputs 10 AWG, A high-side outputs 18 AWG. Final routing/derating/protection remain open.

HT-188000 supplies controller-side leads and CAN plug. Finish free ends; no cockpit-positive bridge into dedicated PD16 wake.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /13 / HALTECH PD16 CONNECTIONS AND CONTROL/
File: 13_PD16_Control.kicad_sch

***Title: 13 / HALTECH PD16 CONNECTIONS AND CONTROL***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-25 | **Rev: J – REVIEW** |
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 13/19 |

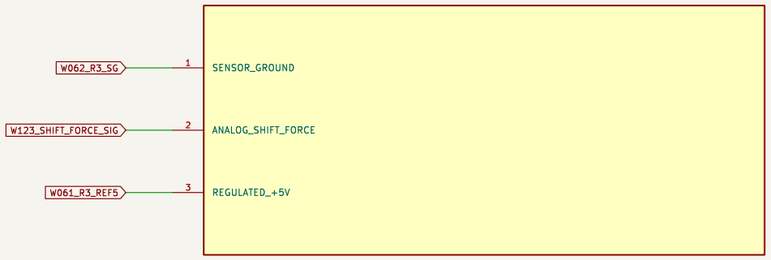
# 14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL

B31 senses driver shift force. B30 (sheet 10) senses actual gear. Both signals are read directly by the Nexus R3.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

B31 | HGT STRAIN–GAUGED KNOB / DT STYLE

HGT shift–force sensing knob




## R3 CONNECTIONS – CONNECTOR C / J11, SHEET 03

Knob cavity 1 –> C26 / sensor signal ground.

Knob cavity 2 –> C28 / AVI11; pull–up disabled.

Knob cavity 3 –> C25 / regulated sensor +5 V.

Internal amplifier: no external amplifier or cut relay.

Shared 5 V load budget: 200 mA maximum at C25.

HGT current demand is unlisted; verify before release.

## HARNESS AND INSTALLATION

Proposed harness extension: three 22 AWG automotive wires; verify contact size and routing.

Use the supplied DT–style cavity numbers; verify the actual mating shell and terminal view.

Wire colors are unverified. Do not assign functions by color. Ground returns to C26 only.

Retain lead strain relief and a flexible service loop through full lever travel.

Mount for fore/aft force. M12 x 1.75 fitment; check adapter against the actual Weddle lever.

## NSP SETUP – REQUIRED BEFORE ENABLING SHIFT CUT

Select Load Cell as flat–shift trigger; assign AVI11. Keep cut/blip disabled during initial checks.

At the selected 5 V supply, HGT resting output is about 1.80 V. Measure rest, pull and push.

Confirm actual signal direction; calibrate trigger voltages and hysteresis for driver force.

Use B30 / AVI10 calibrated gear feedback for closed–loop shift setup; validate every gear.

Tune RPM/TPS enable limits, cut/recovery, maximum cut time and retrigger blockout.

Verify sensor–disconnect behavior. Auto–blip remains disabled until DBW/downshift tuning.

Source: HGT installation guide 63800–IB–SCK–008, 12 July 2024, especially page 5; see README for links.

Pinout applies to HGT pre–terminated DT–style units. Confirm supplied hardware before terminating the mating harness.

HOLD S01: supply–current budget, mating connector, installed signal calibration and tuner validation. No NSP file programmed.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL/
File: 14_Shift_Control.kicad_sch

***Title: 14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL***

| Size: A3 | Date: 2026–10–25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad–cli 7.0.10 | | Id: 14/19 |

# 15 / TRACTION CONTROL AND SPARE I/O

Selected: Delta CAN wheel speeds to R3. Haltech support response supplied by owner; see README / Sources.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

## SELECTED WHEEL-SPEED PATH

Four wheel sensors -> Delta ABS (sheets 08-09).

Delta public J22:3 CAN H / J22:2 CAN L -> R3 CAN2.

R3 J11/C21 = CAN2 H; J11/C22 = CAN2 L (sheet 03).

NSP: select the available Delta ABS vehicle-CAN profile.

Assign FL, FR, RL and RR speed sources to CAN input.

R3 compares calibrated front/rear speeds for traction control.

J22:7-10 pulse outputs stay NC; R3 SPI3-6 stay spare.

Sensor signal/return pairs remain connected to Delta only.

## SUPPORT CONFIRMATION AND CHECKS

Haltech support: CAN is suitable; Delta profile is available.

Gen 2 qualification: verify protocol against installed NSP.

Confirm public-bus bitrate, termination and harness mapping.

Published wheel-speed broadcast: 100Hz / nominal 10ms.

Check wheel identity, units and calibration for all four wheels.

Validate TC enable, slip tuning and sensor/CAN fault response.

Do not infer receive/TC rate from ECU-to-dash broadcasts.

This drawing does not configure NSP or validate live data.

## SPARE CHANNELS - REVISION J

R3: 1 dedicated AVI; 4 SPI; 6 DPO; 0 HBO; 0 HCO.

R3 AVI9/C18 spare (onboard MAP); SPI3-6 remain spare.

Fuel pressure AVI6/C15; intake air temperature AVI8/C17.

PD16: 2 AVI; 4 SPI; 2 HBO; 6 x 8A HCO; 0 x 25A HCO.

PD16 AVI1/A20 engine oil; AVI2/A21 trans; A11 return.

iC-7: 3 spare legacy sensor AVI1-3; AVI4 = fuel level.

NSP additionally exposes AVI5-9, 3 DPI and 4 DPO.

Extended dash capacity is firmware-dependent; see README.

## ADDITIONS - CHANNEL BUDGET

HT-010206 replaces B5 intake-air placeholder: no extra AVI.

HT-010910 replaces B11 fuel-pressure placeholder: no extra AVI.

Two HT-010306 use PD16 AVI1/2: 1k pull-ups ON.

HT-011600 TMS-4 adds CAN receiver; no ECU sensor inputs.

Camera retains A7/HCO5; keypad uses CAN, not R3 SPI.

Keypad camera toggle; proposed button 10 / default OFF.

One O2 uses onboard WBI1; no AVI or external WB unit.

I/O fit is verified; power/reference budgets remain holds.

NSP: PD16 AVI1 = Engine Oil Temperature; AVI2 = Transmission Oil Temperature. Separate HT-010306 calibrations.

iC-7 fuel level: pin 21, sensor return pin 30. NSP sharing minimums: R3 1.26.4 / dash 2.26.3; verify installed versions.

TMS-4 uses CAN1 via supplied adapter (sheet 16). CAN2 remains Delta only; keep two effective terminations per bus.

HOLD T01: Delta Gen 2 profile/bitrate, wheel calibration and traction-control failure behavior before enabling TC.

HOLD F02: combined R3 5V reference current, cockpit supply/return load and camera-kit variant/protection test.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /15 / TRACTION CONTROL AND SPARE I/O/

File: 15_Traction_IO.kicad_sch

***Title: 15 / TRACTION CONTROL AND SPARE I/O***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
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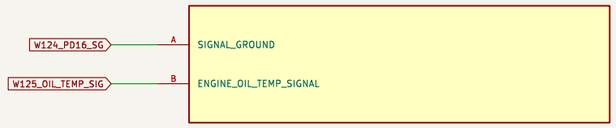
# 16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP

Two HT-010306 high-range 150C sensors use separate PD16 AVIs. HT-011600 tyre sensors communicate wirelessly.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

B32 | HT-010306 / ENGINE OIL

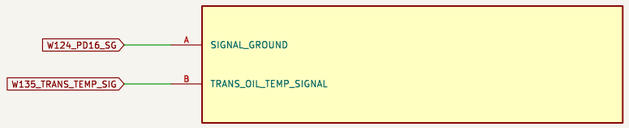
HT-010306 engine oil temperature




Engine: PD16 A20 / AVI1 -> B32:B; A11 -> B32:A.

B33 | HT-010306 / TRANS OIL

HT-010306 transmission oil temperature




Transmission: PD16 A21 / AVI2 -> B33:B; A11 -> B33:A.

Enable each AVI internal 1k pull-up; no external 5V wire.

Both: load Haltech Oil 150C 1/8 NPT HT010306.cal.

Separate engine/trans channels; verify curve and sharing.

EV1 harness view: A return / B signal. A11: sensors only.

## FOUR WHEEL SENSORS / WIRELESS

Fit internal sensors to FL / FR / RL / RR; batteries are inside the sensors. No wires run out to the wheels.

NSP: Connections > CAN > Haltech CAN System > Tyre Monitoring / TMS-4 Enable; assign IDs to actual corners.

TMS-4: verify corner identity, pressure/temperature units, alarm thresholds and lost-sensor response. Mount receiver high in vehicle.

Manufacturer typical in-motion refresh: 60 seconds. TPMS is for tyre monitoring; it is NOT a wheel-speed source for traction control.

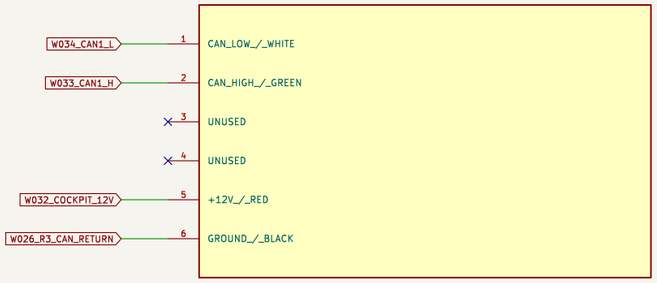
PD16 engine/trans temp: separate open/short diagnostics, valid range, alarms and stale-CAN behavior. Thresholds remain unset.

Confirm trans mounting/immersion, thread clearance, 150C range and calibration against a reference; no mechanical location specified.

Sources: Haltech TMS-4 QSG; HT-010306 info sheet rev 4.2; PD16 diagram rev 1.5. See Sources/ and README.

U23 | HT-011600 / TMS-4 MCON6

Haltech TMS-4 receiver / HT-011600




Supplied DTM4 -> MCON6 adapter:

DTM1 +12V -> MCON5; DTM2 GND -> MCON6.

DTM3 CAN H -> MCON2; DTM4 CAN L -> MCON1.

Receiver: 12-32V; 50mA ambient operating current.

Use supplied adapter; do not infer plug viewing direction.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP/
File: 16_TPMS_Oil_Temp.kicad_sch

***Title: 16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
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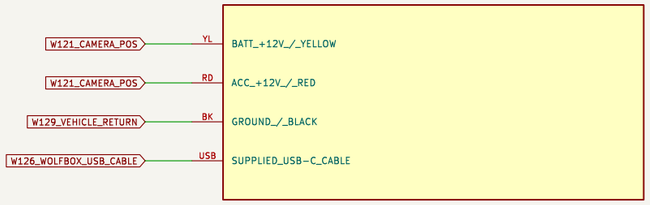
# 17 / WOLFBOX G900 PRO CAMERA POWER

Keypad request –> R3 CAN control –> PD16 A7 HCO5. Complete power removal when commanded OFF; no parking supply.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

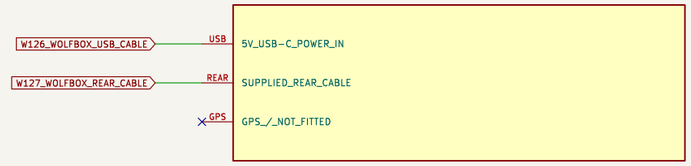
U24 | G900 PRO HARDWIRE KIT

WolfBox G900 Pro hardwire converter




U21 | G900 PRO MIRROR / FRONT CAM

WolfBox G900 Pro mirror and front camera




## SWITCHED–ONLY DESIGN / VERIFY KIT ON RECEIPT

Yellow BATT AND red ACC take the same camera–switched

12V feed from PD16 A7. Black returns to vehicle return.

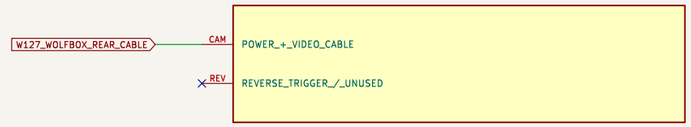
This deliberately removes all camera power when OFF.

Retain the WolfBox converter and its supplied USB–C cable.

Use a kit matching the mirror version/input–current label.

U25 | WOLFBOX REAR CAMERA

WolfBox rear camera




No separate 12V or USB supply goes to the rear camera.

Select normal rear view; disable screen auto–off as needed.

## CONTROL AND PROTECTION

Camera button toggles PD16 A7; proposed keypad position 10, initial OFF. R3 / PD16 / keypad must already be awake.

Use high–side ON/OFF, full supply when ON; do not PWM the converter supply. Verify orderly recording on power loss.

USB and rear–cable W labels denote complete vendor cable assemblies, NOT single wires or USB cavity numbers.

HOLD C01: confirm purchased kit labels/variant and switched–only operation with both BATT/ACC on A7 before installation.

3A/4A at 5V are mirror supply ratings, not measured continuous draw. At 12V / 90% efficiency: about 1.4A / 1.9A input.

Select PD16 trip/inrush/retry settings to protect actual kit leads; retain vendor inline protection if fitted. Do not use 8A blindly.

No raw–battery parking feed; camera stays downstream of master isolation. GPS and reverse lead omitted as the simple rear–view basis.

Insulate unused reverse lead. Verify boot time, rear view, CAN–loss OFF, cranking and master–stop behavior on actual hardware.

DRAFT – NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /17 / WOLFBOX G900 PRO CAMERA POWER/

File: 17_Wolfbox_Camera.kicad_sch

***Title: 17 / WOLFBOX G900 PRO CAMERA POWER***

| Size: A3 | Date: 2026–10–25 | **Rev: J – REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad–cli 7.0.10 | | Id: 17/19 |

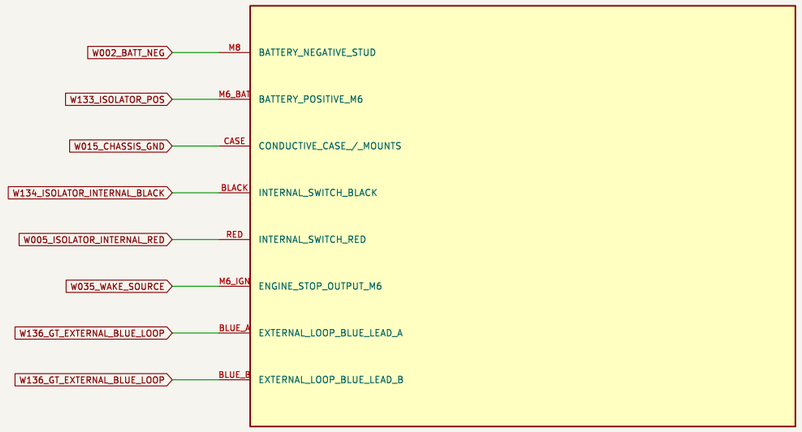
# 18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS

GT manual rev1.0 / 07Nov2025, method 1 ECU/PDM input. ONE internal stop switch; no external switch. Blocks are proposed parts.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

U2 | CARTEK GT / VERIFY s OR x

## CARTEK GT battery isolator



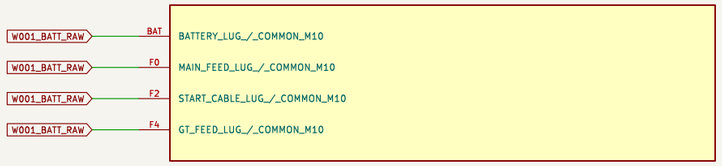

F4 | GT CONTROL / 5A PROPOSED

## GT CONTROL / 5A PROPOSED




TB1 | FS46540 / RAW + / PROPOSED

## Sierra FS46540 / PROPOSED



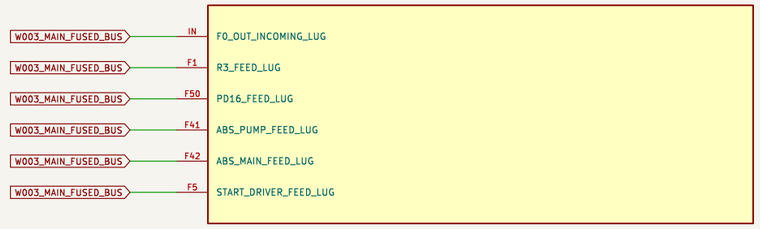

S40 | B0ONTT91Y0 / LATCHING NC

## BAOMAIN latching stop / NC




TB2 | SUM-900533 / FUSED + / PROPOSED

## Summit SUM-900533 / PROPOSED




S40: one cockpit stop button, reachable inside/outside.

Reset/released = NC closed / ON; pressed = latched open / OFF.

Verify actual NC pair. Pins 1/2 are functional, NOT cavity IDs.

Join BLUE_A to BLUE_B permanently; no chassis connection. S41 removed. No battery voltage on switch leads.

GT M6_IGN -> S1 -> R3 A13 only. K1 removed; F0 -> TB2 direct. GT output is fixed 40A protected.

TB1: four lugs / common M10 post, stack and covers TBD. TB2: six lugs / four 3/8 studs; allocation TBD.

TB2 listing conflicts: specific 250A vs family 150A. Confirm actual rating, thermal duty and permitted lug stacks.

GT: 7-18V; 600A cranking / 2000A surge. Confirm received GT-s/GT-x, terminal labels and actual starter current.

Install in manual order: chassis bond, switch/blue loop, engine-stop, M6 battery positive, then M8 negative. Limit temp to +85C pending source conflict.

F4 5A remains PROPOSED for the GT/control feed. Size wiring/protection for actual GT + R3 A13 load, not a contactor coil.

K1 deleted by owner. Verify R3 A13 OFF stops engine/fuel and PD16 outputs promptly before GT battery isolation.

Twist reset may re-arm power; require a fresh start command. Charger/booster cables both at battery posts; no negative bypass.

DRAFT - NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS/

File: 18_Master_Stop.kicad_sch

***Title: 18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS***

| Size: A3 | Date: 2026-10-25 | **Rev: J - REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 18/19 |

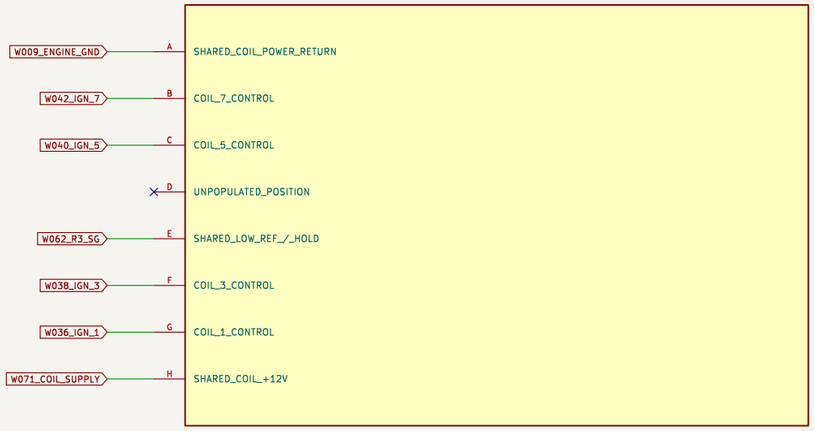
# 19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS

Retained donor wiring / 7 populated bank ways / related-year GM cavity reference; confirm actual donor continuity and orientation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

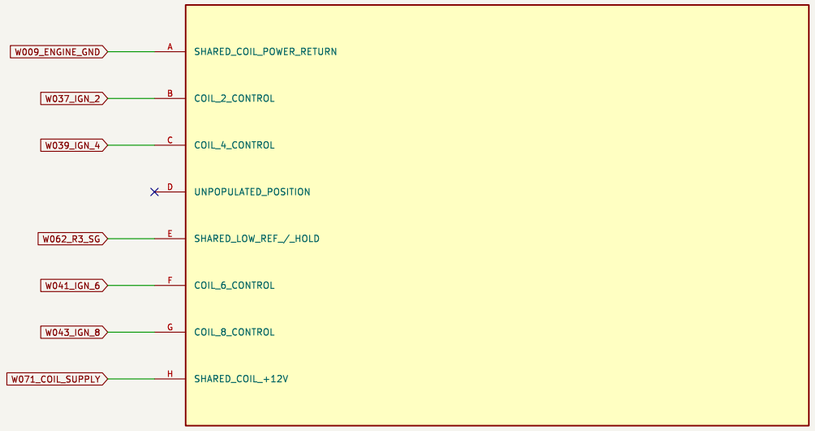
X111 | OEM L94 / ODD / 7-WAY

Stock L94 odd-bank coil connector




X113 | OEM L94 / EVEN / 7-WAY

Stock L94 even-bank coil connector




## OEM COIL BANK LOOMS — RETAIN THEIR INTERNAL SPLICES

Each bank: four triggers plus common +12V, power ground and low reference. D is unavailable, not an eighth live wire.

Reference coil plug: A=power ground, B=low reference, C=trigger, D=+12V. Device IDs on sheet 04 stay functional.

S paths in the continuous views are RETAINED STOCK wiring. C paths are new integration wires; no new coil-branch cuts assumed.

## DONOR INJECTOR WIRING — REUSE WITH LS3 EV6 / USCAR INJECTORS

Retain eight individual driver legs and separate odd/even +12V branches. No factory injector bulkhead plug is assumed.

CUT-INJ labels mark functional donor cut/termination boundaries, NOT connector cavities; locations/join hardware TBD.

Factory feed circuits may also serve coils: isolate retained injector feeds before HCO1. NEVER connect HCO1 to HCO2.

HOLD: verify LS3 injector PN, plug key/latch, connector orientation, harness reach, condition, continuity and calibration.

GROUND HOLD: bank A -> engine; bank E -> R3 C26 is retained pending exact coil evidence. No W009/W062 bridge permitted.

GM separates low reference; Haltech generic Gen IV guide also advises both coil grounds at the head. Resolve with exact coil/R3 guidance.

Reference: Stock_Engine_Harnesses.json and OEM_Harness_Interfaces.tsv. Source cavities/colors are not inspected donor measurements.

No downstream bank fuses restored. HCO1 injector / HCO2 coil trip coordination with every OEM branch and contact remains OPEN.

DRAFT — NOT A BUILD RELEASE

**Danny's Race Car | Harness design**

Sheet: /19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS/
File: 19_Stock_Engine_Harnesses.kicad_sch

***Title: 19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS***

| Size: A3 | Date: 2026-10-25 | **Rev: J — REVIEW** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.10 | | Id: 19/19 |